Applied Digital Logic Design(ECE 101)
Multiple Choice Questions
(2025-26 Odd Semester)
Unit 1

Sl. No.	Question	Answer
1.	Which of the following is true about analog signals? A. They have discrete values B. They represent data using binary numbers C. They vary continuously over time D. They are only used in computers	**C**
2.	Digital signals are preferred over analog signals in computers because: A. They are continuous B. They are immune to noise C. They are slower D. They use infinite levels of voltage	**B**
3.	The binary equivalent of the decimal number 13 is: A. 1011 B. 1101 C. 1001 D. 1110	**B**
4.	Which of the following is a valid octal number? A. 178 B. 88 C. 725 D. 91	**C**
5.	The hexadecimal number 1A is equivalent to the decimal: A. 26 B. 27 C. 18 D. 16	**A**
6.	What is the result of 1011 + 1101 in binary? A. 10111 B. 11000 C. 11100 D. 10000	**B**
7.	What is 1101 - 101 in binary? A. 1000 B. 1010 C. 1110 D. 1001	**A**
8.	Which logic gate gives a HIGH output only when all inputs are HIGH? A. OR B. NAND C. AND D. XOR	**C**
9.	The output of a NOT gate is: A. Always 1 B. The same as the input C. The complement of the input D. 0 if input is 0	**C**

10. According to Boolean algebra, A + A = ? **B**
A. 0
B. A
C. 1
D. A'

11. A don’t care condition (X) can be treated as: **C**
A. Always 0
B. Always 1
C. Either 0 or 1, based on simplification
D. Eliminated

12. A completely specified function means: **B**
A. Don’t care conditions are present
B. All input combinations have defined output
C. Some outputs are unknown
D. No inputs are present

13. Incompletely specified Boolean functions are represented using: **B**
A. Only minterms
B. Don’t care conditions
C. Binary only
D. SOP form only

14. What is the binary equivalent of the octal number 17? **C**
A. 1111
B. 11001
C. 10001
D. 1000

15. Which logic gate gives a LOW output only when all inputs are HIGH? **A**
A. NAND
B. NOR
C. XOR
D. XNOR

16. “Don’t care” conditions in Boolean expressions are used for: **B**
A. Increasing circuit cost
B. Optimising the logic function
C. Removing minterms
D. Adding extra inputs

17. Convert the hexadecimal number 3F.A to its binary equivalent: **A**
A. 111111.101
B. 111111.110
C. 111110.101
D. 111111.111

18. The output of the logic circuit with inputs A, B, and C connected to a NAND gate, followed by a NOT gate, is equivalent to: **B**
A. A + B + C
B. ABC
C. $A \oplus B \oplus C$
D. A' + B' + C'

19. What is the hexadecimal equivalent of the binary number 11110110.1101? **A**
A. F6.D
B. E7.6
C. D7.1

D. F5.C

20. What is the number of minterms in a 4-variable Boolean function? **C**
A. 4
B. 8
C. 16
D. 32

Unit 2

1. Simplify the expression using Boolean algebra: **A**
AB + A'B + AB'
A. A + B
B. B' + A'
C. A ⊕ B
D. A + B'

2. What is the simplified form of the Boolean expression:A + AB **B**
A. B
B. A
C. AB
D. A + B

3. Which of the following is a valid group in a K-map? **C**
A. 3 adjacent 1s
B. Diagonal grouping
C. 2, 4, or 8 adjacent 1s
D. Random 1s

4. What is the simplified expression from a 3-variable K-map for A,B, C with minterms 1, 3, 5, and 7? **C**
A. A'B'
B. B
C. C
D. B + C

5. The Quine-McCluskey method is also known as: **B**
A. Algebraic method
B. Tabular method
C. Karnaugh method
D. DeMorgan method

6. The function F = A' + B can be implemented using: **A**
A. NOT and OR gates
B. AND and NOT gates
C. NOR only
D. NAND only

7. What is the complement of the Boolean expression A + B? **C**
A. A' + B'
B. AB
C. A'B'
D. A + B'

8. A 4-variable K-map has how many cells? **B**
A. 8
B. 16
C. 32
D. 64

9. Which of the following is not a valid group in a 4-variable K-map? **C**
A. 1 cell
B. 2 cells
C. 5 cells
D. 8 cells

10. In Quine-McCluskey, a table entry with two '-'s represents: **C**
A. One minterm
B. Two minterms
C. Four minterms
D. Don’t care term

11. Universal gates can be used to build: **A**
A. Any logic circuit
B. Only SOP forms
C. Only combinational logic
D. Only arithmetic circuits

12. Which of the following statements is true regarding the function F=AB+A’B+AB’+A'B'. **A**
A. F is always equal to 1
B. F = A + B
C. F simplifies to A + B
D. F simplifies to A $\oplus$ B

13. The circuit implementing F=A’B+AB’ is called: **C**
A. XNOR
B. NOR
C. XOR
D. NAND

14. In the Quine-McCluskey method, what is the significance of a term that covers both minterm 5 (0101) and minterm 7 (0111)? **C**
A. Difference in 2 bits
B. Invalid pairing
C. Valid pairing with one variable eliminated
D. Group with don’t care

15. A group of eight 1s in a 4-variable K-map represents: **A**
A. A single literal
B. A product term with two literals
C. A don’t care
D. A constant 1

16. A function F(A,B,C) has minterms 0, 1, 2, 5, and don’t-care terms 3, 7. What is the minimized expression using K-map? **D**
A. A’C’+B
B. A’+B
C. A’C+B’C’
D. A’+C

17. In the Quine-McCluskey method, the initial grouping is based on: **C**
A. Output values
B. Number of variables
C. Number of 1’s in the binary representation of minterms
D. Prime implicants

18. The Quine-McCluskey method is primarily used for: **B**
A. Converting binary to decimal
B. Simplifying Boolean expressions

C. Designing flip-flops
D. Building combinational circuits

19. What is the output of the Boolean function F = A.(B + C) when A = 1, B = 0, and C = 1? **B**
A. 0
B. 1
C. Undefined
D. Cannot be determined

20. Which of the following is true about group formation in K-maps? **C**
A. Groups must include diagonal cells
B. Groups can include only 2 minterms
C. Groups must be powers of 2
D. Groups can overlap only once

Unit 3

1. Combinational circuits are defined as those whose output: **B**
A. Depends on both present and past inputs
B. Depends only on current inputs
C. Depends only on clock pulses
D. Depends on feedback

2. A combinational circuit has: **C**
A. Memory elements
B. Feedback paths
C. No memory and no feedback
D. Clock signal to control output

3. The basic building blocks of combinational logic include: **C**
A. Flip-flops
B. Latches
C. Logic gates
D. Registers

4. A half adder has how many inputs and outputs? **B**
A. 2 inputs, 1 output
B. 2 inputs, 2 outputs
C. 1 input, 2 outputs
D. 3 inputs, 1 output

5. The output expressions for a half adder are: **A**
A. Sum = A XOR B; Carry = A AND B
B. Sum = A OR B; Carry = A AND B
C. Sum = A XOR B; Carry = A OR B
D. Sum = A AND B; Carry = A OR B

6. Which gate combination best describes the carry-out of a full adder? **A**
A. AB + BC + AC
B. $A \oplus B$
C. $A \oplus B \oplus Cin$
D. A + B + C

7. A 4-bit parallel adder adds: **B**
A. 4 bits to a 1-bit number
B. 4 bits + 4 bits
C. 2 bits at a time
D. 1 bit per clock

8. A half subtractor produces: **B**
A. Difference and Sum
B. Difference and Borrow
C. Carry and Sum
D. Borrow and Carry

9. In a half subtractor, the borrow output is: **A**
A. A'B
B. AB'
C. A XOR B
D. A AND B

10. Which of the following is not an output of a comparator? **D**
A. A = B
B. A > B
C. A < B
D. A XOR B

11. A 4-bit comparator compares: **B**
A. One 4-bit input with 1-bit
B. Two 4-bit binary numbers
C. Four 1-bit numbers
D. Two decimal numbers

12. A 3-to-8 line decoder has: **A**
A. 3 inputs and 8 outputs
B. 8 inputs and 3 outputs
C. 3 inputs and 3 outputs
D. 8 inputs and 1 output

13. An 8-to-3 encoder has: **A**
A. 8 inputs, 3 outputs
B. 3 inputs, 8 outputs
C. 3 inputs, 3 outputs
D. 2 outputs, 4 inputs

14. A multiplexer: **A**
A. Selects one input from many inputs
B. Sends multiple outputs to one input
C. Adds two binary numbers
D. Compares two numbers

15. A 4-to-1 multiplexer has how many select lines? **B**
A. 1
B. 2
C. 3
D. 4

16. The number of input lines in an 8:1 multiplexer is: **A**
A. 8
B. 3
C. 4
D. 2

17. A multiplexer works as a: **B**
A. Data distributor
B. Data selector
C. Decoder
D. Comparator

18. A 16-to-1 multiplexer has how many select lines? **B**
A. 3
B. 4
C. 5
D. 2

19. A multiplexer can be used to implement: **C**
A. Latches
B. Flip-flops
C. Any combinational function
D. Only parity generator

20. The output of a half subtractor when A = 0 and B = 1 is: **A**
A. Diff = 1, Borrow = 1
B. Diff = 0, Borrow = 1
C. Diff = 1, Borrow = 0
D. Diff = 0, Borrow = 0

Unit 4

1. A bistable circuit can be used as: **C**
A. Decoder
B. Counter
C. Memory element
D. Comparator

2. The main difference between a latch and a flip-flop is: **A**
A. Clock dependency
B. Output type
C. Number of inputs
D. Number of states

3. In an SR latch made from NOR gates, if S = 1 and R = 0, the output is: **A**
A. Q = 1, Q' = 0
B. Q = 0, Q' = 1
C. Q = Q' = 1
D. Q = Q' = 0

4. What is the characteristic equation of the SR flip-flop? **C**
A. $Q^+ = R + S$
B. $Q^+ = Q$
C. $Q^+ = S + R'Q$
D. $Q^+ = S + R'$

5. The purpose of the master-slave configuration is to: B
A. Reduce cost
B. Prevent the race-around condition
C. Increase speed
D. Increase invalid states

6. Which equation correctly describes the next state of a T flip-flop? **C**
A. $Q^+ = Q$
B. $Q^+ = D$
C. $Q^+ = T \oplus Q$
D. $Q^+ = T + Q$

7. If T = 1 in a T flip-flop, what is the next output state? **D**
A. Set
B. Reset

C. No change
D. Toggle

8. The expression $Q^+ = Q$ indicates that: **B**
A. The flip-flop toggles
B. The output remains unchanged
C. The flip-flop resets
D. The flip-flop sets

9. In a JK flip-flop, if J and K inputs remain high for a long duration, what will happen? **B**
A. Flip-flop will latch
B. Flip-flop will toggle continuously
C. Outputs freeze
D. Nothing

10. From the characteristic equation of a JK flip-flop, deduce Q^+ if Q = 1, J = 1, K = 0. **A**
A. 1
B. 0
C. Toggle
D. Invalid

11. What is the output of a JK flip-flop with J = 0, K = 1, and initial state Q = 1? **A**
A. 0
B. 1
C. Toggles
D. Undefined

12. The JK flip-flop behaves like a T flip-flop when: **B**
A. J = 1, K = 0
B. J = K = 1
C. J = 0, K = 1
D. J = K = 0

13. To convert a SR flip-flop to a D flip-flop, connect: **A**
A. S = D, R = D'
B. S = D, R = 0
C. S = D', R = D
D. S= 1, R= 1

14. The characteristic equation of a D flip-flop is: **A**
A. $Q^+ = D$
B. $Q^+ = T \oplus Q$
C. $Q^+ = JQ' + K'Q$
D. $Q^+ = S + R'$

15. A bistable element is a circuit that has: **B**
A. One stable state
B. Two stable states
C. No stable state
D. Multiple outputs

16. A D flip-flop eliminates the invalid state of: **A**
A. SR flip-flop
B. JK flip-flop
C. T flip-flop
D. Master-slave

17. The slave in a master-slave JK flip-flop is enabled when: **B**
A. The clock is HIGH

B. The Clock is LOW
C. The clock is toggled
D. The clock is stable

18. If T = 0 in a T flip-flop, what is the next output state? **C**
A. Set
B. Reset
C. No change
D. Toggle

19. In an SR latch made from NOR gates, if S = 0 and R = 1, the output is: B
A. Q = 1, Q' = 0
B. Q = 0, Q' = 1
C. Q = Q' = 1
D. Q = Q' = 0

20. JK flip-flops are preferred over SR flip-flops because: C
A. They are cheaper
B. They eliminate race conditions
C. They eliminate the invalid state
D. They are faster

Unit 5

1. A 4-bit shift register using D flip-flops can store: **C**
A. 2 bits
B. 8 bits
C. 4 bits
D. 16 bits

2. In a serial-in serial-out (SISO) shift register using D flip-flops, data is: **B**
A. Loaded in parallel and retrieved in serial
B. Loaded in serial and retrieved in serial
C. Loaded in parallel and retrieved in parallel
D. Randomly loaded and retrieved

3. In a right shift register using D flip-flops, data moves: **B**
A. From right to left
B. From left to right
C. Bi-directionally
D. Randomly

4. A synchronous MOD-16 counter uses how many flip-flops? **A**
A. 4
B. 3
C. 5
D. 6

5. A MOD-5 synchronous counter cycles through how many states? **D**
A. 4
B. 10
C. 6
D. 5

6. Synchronous counter design often uses Karnaugh maps to: **C**
A. Store logic
B. Identify hazards
C. Simplify D/T input logic
D. Generate toggles

7. A synchronous counter designed using D flip-flops counts 000 → 001 → 010 → 011. This is: **B**
   A. MOD-3
   B. MOD-4
   C. MOD-2
   D. Ring counter
8. For a MOD-6 counter using T flip-flops, how many flip-flops are required? **B**
   A. 2
   B. 3
   C. 4
   D. 6
9. Which is true about synchronous counters? **C**
   A. Trigger on random edges
   B. No timing control
   C. All flip-flops are triggered at the same time
   D. Require fewer gates than ripple
10. For a synchronous 4-bit up-counter using T flip-flops, the T inputs should be: **B**
    A. All 1
    B. $T_0 = 1$, $T_1 = Q_0$, $T_2 = Q_0 \cdot Q_1$, $T_3 = Q_0 \cdot Q_1 \cdot Q_2$
    C. T = Q
    D. Tied to ground
11. In a synchronous counter, all flip-flops are triggered by: **D**
    A. Their Q output
    B. Feedback logic
    C. A ring oscillator
    D. The same clock signal
12. A 3-bit ripple down-counter using T flip-flops starts at 111. Next state? **B**
    A. 000
    B. 110
    C. 001
    D. 011
13. What is the T input for each T flip-flop in a binary ripple counter? **A**
    A. 1
    B. 0
    C. Tied to Q of the previous
    D. Alternating 1 and 0
14. In a ripple counter, each flip-flop is triggered by: **C**
    A. A global clock
    B. Its clock
    C. The output of the previous flip-flop
    D. Tied to logic 1
15. In an asynchronous 3-bit up-counter using T flip-flops, how many states are possible? **C**
    A. 4
    B. 6
    C. 8
    D. 16
16. An asynchronous counter is also called: **A**
    A. Ripple counter
    B. Ring counter

C. Binary counter
D. Modulo counter

17. Which type of shift register can perform both left and right shifts? **D**
A. SISO
B. PIPO
C. SIPO
D. Bidirectional

18. A parallel-in, parallel-out (PIPO) shift register behaves like: **A**
A. Memory
B. Counter
C. Encoder
D. Demux

19. The output of each D flip-flop in a shift register is connected to: **B**
A. The clock input of the next
B. The D input of the next
C. The Q' of the previous
D. Ground

20. In a shift register, which flip-flop is best suited? **B**
A. T
B. D
C. SR
D. JK